Calibration Strategy for the Quantitative Model

Provisional note

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1 Objective

This note lays out the proposed calibration of the quantitative model. Its purpose is to make the empirical content of each estimated parameter explicit. The calibration is organized in economic blocks. Within each block, parameters are associated with moments that isolate the variation governed most directly by those parameters. All parameters are ultimately chosen jointly, but the target system is designed so that each parameter has one primary empirical counterpart, or belongs to a stated two-parameter, two-moment block.

The revised calibration has four blocks: saving and bequests; baseline consumption and housing demand; child costs and fertility; and tenure and housing supply. Table 2 gives the complete mapping. The remainder of the note explains the economic content of each block and records each moment's construction and current status.

2 Parameters assigned outside the estimation

Table 1 records the maintained inputs inherited from the current quantitative specification. These parameters describe timing, survival, income risk, housing finance, the available housing products, and the elasticity of housing supply. They are not counted among the fourteen internally estimated parameters.

Table 1: Parameters assigned outside the estimation

Object	Value	Basis
Model period; entry, retirement, maximum age	4 years; 18, 66, 85	Model timing
Post-retirement survival	Age-specific	2023 Social Security life table
Expected dependent-child duration	18 years	Child-maturity restriction
Risk aversion σ	2	Maintained preference curvature
Annual real return	2.0%	Long-run real rate
Annual depreciation; property tax	1.1%; 1.0%	Housing user cost
Financed share; sale cost	0.80; 6%	Standard housing-finance values
Annual income persistence; innovation s.d.	0.960; 0.0645	Current provisional income process
Owner sizes; largest rental unit	$\{2, 4, 6, 8, 10\}$; 6 rooms	ACS room distributions
Housing-supply elasticity η	1.75	Saiz (2010)
Entrant wealth distribution	Ages 18–24	PSID childless renters
Child shifter in bequest utility	$\theta_n = 0$	Maintained restriction

The income process remains provisional. Revising income risk changes precautionary saving and therefore requires re-estimating the saving, consumption, tenure, and bequest blocks. It cannot be changed after calibration while retaining the parameter estimates reported under the current process.

3 Saving and bequests

The saving and bequest block follows De Nardi and Yang (2014) as closely as the housing environment permits. Bequeathable wealth includes both liquid wealth and the gross value of owner-occupied housing:

$$W^B = b + Ph. \tag{1}$$

The warm-glow bequest function is

$$\mathcal{B}(W^B) = \theta_0 \, \frac{(\theta_1 + \max\{W^B, 0\})^{1-\sigma} - \theta_1^{1-\sigma}}{1-\sigma}. \tag{2}$$

The subtraction normalizes $\mathcal{B}(0) = 0$ and does not change marginal bequest utility. For positive estates,

$$\mathcal{B}'(W^B) = \theta_0(\theta_1 + W^B)^{-\sigma}. \tag{3}$$

Thus θ_0 controls the scale of the bequest motive, while θ_1 controls how the motive varies across the estate distribution. A larger shift makes small estates less attractive relative to large estates and therefore governs the extent to which bequests behave as a luxury good.

The three parameters $(\beta, \theta_0, \theta_1)$ are associated with three moments:

$$\beta \longleftrightarrow \frac{\text{aggregate household wealth}}{\text{aggregate after-tax earnings}}, \tag{4}$$

$$\theta_0 \longleftrightarrow \frac{\text{annual bequest flow}}{\text{aggregate household wealth}}, \tag{5}$$

$$\theta_1 \longleftrightarrow \frac{p_{90}(\text{bequest at death})}{\text{annual household income}}. \tag{6}$$

The first moment disciplines total lifecycle wealth accumulation. The second isolates the level of resources intentionally carried to death. The third disciplines the upper tail of estates and hence the luxury component of the bequest motive. De Nardi and Yang (2014) use values 6.90, 0.0088, and 4.53 for these three objects. The initial implementation will reproduce their definitions. Modern target values will replace the original values only after the underlying numerator, denominator, population, and price-year conventions have been reconstructed consistently.

This block replaces the current practice of using young liquid wealth, a late-life estate median, and an old-age nonhousing-wealth threshold to jointly discipline patience and bequests. Those moments remain useful validation outcomes, but they do not separately measure average wealth accumulation, aggregate bequest intensity, and estate curvature.

4 Consumption and housing demand

Preferences are CRRA over a Stone–Geary composite. For a renter with no dependent children, the within-period composite is

$$(c - \bar{c}_0)^{\alpha_c}(h^R - \bar{h}_0)^{1-\alpha_c}. \tag{7}$$

For an interior renter with allocatable expenditure $x = c + rh^R$, conditional housing expenditure satisfies

$$rh^R = -(1-\alpha_c)\bar{c}_0 + (1-\alpha_c)x + \alpha_c r\bar{h}_0. \tag{8}$$

This provides a direct three-parameter mapping. In the auxiliary regression

$$rh^R = a + bx + dr + u, \tag{9}$$

the primitives are recovered as

$$\alpha_c = 1-b, \qquad \bar{h}_0 = \frac{d}{1-b}, \qquad \bar{c}_0 = -\frac{a}{b}. \tag{10}$$

We estimate this system on the 2019–2023 CEX Interview files for cash renters ages 30–55 with no dependent children. The local rent per room is constructed from the opposite half of a deterministic consumer-unit split within public CEX geography, so a household's own rent and rooms do not mechanically determine its assigned unit price. Expenditures are converted to 2023 dollars, household income uses the BLS collected-or-imputed measure, repeated interviews are clustered by consumer unit, and the regression uses the CEX final weight. The pooled estimates imply

$$\widehat{\alpha}_c = 0.733, \qquad \widehat{\bar{h}}_0 = 2.630, \qquad \widehat{\bar{c}}_0 = 0.397 \quad \text{(four-year model units).} \tag{11}$$

The corresponding one-year estimates are 0.724, 2.672, and 0.440, so the result is not driven by the 2023 cross section. Restricting the pooled sample to complete income reporters gives 0.735, 2.646, and 0.395. The ACS mean rooms of 3.805 remains a validation outcome: it is the total quantity chosen, whereas $\bar{h}_0$ is the minimum housing term recovered from the demand system.

5 Children and family size

Dependent children shift the household's committed bundle according to

$$\bar{c}(n,s) = \bar{c}_0 + \bar{c}_n n_s, \tag{12}$$
$$\bar{h}(n,s) = \bar{h}_0 + \bar{h}_{\text{jump}}\mathbf{1}\{n_s \geq 1\} + \bar{h}_n n_s. \tag{13}$$

The consumption cost per child is disciplined by the change in nondurable consumption expenditure associated with an additional dependent child, conditional on household resources:

$$\bar{c}_n \longleftrightarrow \frac{\partial E[c \mid y, b, h, a]}{\partial n_s}. \tag{14}$$

The pooled 2019–2023 CEX estimate, controlling for income, age, adult household size, tenure, rooms, and interview quarter, is 1,289 in annual 2023 dollars per dependent child. Relative to weighted mean annual income and the model's four-year period, this gives a provisional target $\bar{c}_n = 0.0437$. Restricting the sample to complete income reporters gives 0.0443. The estimate lies just below the lower edge of the current search region and should therefore enter with an explicit sensitivity analysis. Fertility and ownership differences by family size are not used as substitutes for this direct consumption-cost moment.

The two housing-need parameters have direct physical counterparts:

$$\bar{h}_{\text{jump}} \longleftrightarrow \text{change in rooms following the first child,} \tag{15}$$
$$\bar{h}_n \longleftrightarrow E[h \mid 3+ \text{ children}] - E[h \mid 1\text{–}2 \text{ children}]. \tag{16}$$

The first-child event-study moment and the larger-family rooms gap are already available. Together they distinguish the extensive housing adjustment associated with becoming a parent from the incremental space requirement of additional children.

The direct utility from children and the dispersion in family-size choices form a two-parameter, two-moment block:

$$(\psi_{\text{child}}, \kappa_F) \longleftrightarrow (\text{completed fertility}, \text{childlessness rate}). \tag{17}$$

The level of completed fertility primarily disciplines the average utility value of children. Conditional on that level, childlessness disciplines heterogeneity in family-size choices. Both moments are already constructed from the CPS.

6 Tenure and housing supply

The owner service premium χ_O shifts the average attractiveness of ownership and is associated with the prime-age ownership rate:

$$\chi_O \longleftrightarrow \Pr(\text{owner} \mid \text{ages } 30\text{–}55). \tag{18}$$

The tenure-choice dispersion κ_T determines how sharply otherwise similar households sort into renting and owning. It is associated with residual variation in ownership four years ahead after conditioning on the observable states that the model contains:

$$\kappa_T \longleftrightarrow E\left[\left(O_{t+4} - \widehat{\Pr}(O_{t+4} = 1 \mid X_t)\right)^2\right], \tag{19}$$

where X_t contains current tenure, liquid financial wealth relative to income, income, age, children, marital status, and year. A five-fold person-level cross-fitted logit on the PSID gives 0.1171 (conditional person-bootstrap standard error 0.0021). The identical auxiliary regression must be run on simulated model histories. The ownership–wealth gradient and the unconditional switching rate remain validation moments; they combine residual tenure taste dispersion with observed wealth heterogeneity and lifecycle transitions.

Housing supply is

$$H^S(r) = \bar{H}\left(\frac{r}{\bar{r}}\right)^{\eta}, \tag{20}$$

with η assigned externally. The supply scale is associated with aggregate occupied rooms:

$$\bar{H} \longleftrightarrow E[h \mid \text{ages } 18\text{–}85]. \tag{21}$$

The equilibrium user cost adjusts so that aggregate renter and owner housing demand equals this supply.

7 Complete mapping and implementation

Table 2 records the proposed mapping for all fourteen internally estimated parameters. Every parameter now has a numerical empirical counterpart, although the CEX child-cost estimate and the De Nardi bequest targets remain provisional pending the sensitivity exercises described below.

The numerical target ledger is now complete. Before launching the new calibration, three implementation tasks remain. First, the model must report the same auxiliary LES coefficients and the same

Table 2: Proposed parameter–moment mapping

Parameter	Primary empirical counterpart	Target	Status
β	Aggregate wealth / after-tax earnings	6.90	Published definition
θ_0	Annual bequest flow / aggregate wealth	0.0088	Published definition
θ_1	p_{90} bequest / annual income	4.53	Published definition
α_c	Childless-renter LES expenditure coefficient	0.733	CEX 2019–2023
$\bar{c}_0$	Childless-renter LES intercept	0.397	CEX 2019–2023
$\bar{h}_0$	Childless-renter LES price coefficient	2.630	CEX 2019–2023
$\bar{c}_n$	Nondurable consumption increment per dependent child	0.0437	CEX 2019–2023
$\bar{h}_{\text{jump}}$	First-child rooms response	0.664	Available
$\bar{h}_n$	Rooms gap: 3+ versus 1–2 children	0.368	Available
ψ_{child}	Completed-fertility equivalent	1.918	Available
κ_F	Completed childlessness	0.188	Available
χ_O	Prime-age ownership rate	0.575	Available
κ_T	Cross-fitted four-year residual tenure variance	0.1171	PSID, constructed
$\bar{H}$	Aggregate occupied rooms	5.780	Available

four-year residual-tenure Brier score as the data scripts. Second, the CEX child-cost target is stable to the complete-reporter restriction, but its reduced-form interpretation should remain explicit until the planned equivalence-scale exercise replaces it. Third, the published De Nardi ratios must either be retained explicitly as borrowed targets or reconstructed with modern data; they should not be described as newly measured project moments. These tasks affect measurement and model reporting, not the count of identifying moments.

The current family ownership gap, large-home share among childless owners, owner–renter rooms gap, ownership rates at young and old ages, young liquid wealth, late-life estate median, and old-age nonhousing-wealth share remain in the reporting table as overidentifying outcomes. They evaluate whether the calibrated mechanisms reproduce lifecycle and family differences that were not used to select the corresponding parameters.

Table 3: Disposition of the current target moments

Current moment	Revised role	Parameter or use
Completed fertility	Primary target	ψ_{child}
Childlessness	Primary target	κ_F conditional on fertility
Ownership, ages 30–55	Primary target	χ_O
Family ownership gap	Validation	Family housing–tenure interaction
First-child rooms increment	Primary target	$\bar{h}_{\text{jump}}$
Childless-renter mean rooms	Validation	Baseline housing demand
Childless-owner share in 6+ rooms	Validation	Owner menu and χ_O
Young childless-renter liquid wealth	Validation	Lifecycle saving
Owner–renter rooms gap	Validation	Tenure-specific housing demand
Ownership, ages 25–34	Validation	Early ownership path
Rooms gap: 3+ versus 1–2 children	Primary target	$\bar{h}_n$
Late-life estate median	Validation	Bequest level and lifecycle saving
Old nonhousing wealth above annual income	Validation	Portfolio composition
Ownership, ages 65–75	Validation	Late-life tenure exit
Aggregate occupied rooms	Primary target	$\bar{H}$

References

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